

In this paper, we observe that typical manifolds,

We promise that: equivalent relation on  $X$  generated by: condition means the smallest equivalent relation on  $X$  satisfying the condition.

Rigorously,  $\sim := \bigcap R_i$  where  $R_i$  is a relation on  $X$  satisfying the condition

And,  $I = [0, 1]$ , the subspace of the euclidean space  $\mathbb{R}$ .

$$\begin{array}{ccc} X & & \\ \pi \downarrow & \searrow f & \\ X/\sim & \xrightarrow{\tilde{f}} & Y \end{array}$$

# Cylinder

## Def.

Define an equivalence relation  $\sim$  on  $I \times I$  generated by:  $\forall y \in I, (0, y) \sim (1, y)$   
 The quotient space  $I^2/\sim$  is called the Cylinder.

Prop. The cylinder  $I^2/\sim$  is Homeomorphic to  $S^1 \times I \subseteq \mathbb{C} \times \mathbb{R}$ .

Proof. Define a map

$$f: I^2 \rightarrow S^1 \times I: (x, y) \mapsto (e^{2\pi i x}, y)$$

$$\begin{array}{ccc} I^2 & & f \\ & \searrow & \\ \pi \downarrow & & S^1 \times I \\ I^2/\sim & \xrightarrow{\tilde{f}} & \end{array}$$

Then,  $f$  is continuous onto closed map, because

1. Continuous

$(x, y) \mapsto e^{2\pi i x}$  and  $(x, y) \mapsto y$  are continuous, so  $f$  is also continuous.

2. Onto

Given  $(e^{i\theta}, b) \in S^1 \times I$  where  $\theta \in [0, 2\pi]$ , take  $x = \frac{\theta}{2\pi}$  and  $y = b$ , then clear.

3. Closed

Given closed set  $C \subseteq I^2$ ,  $C$  is compact since  $I^2$  is compact.

So, the continuous image  $f[C]$  is also compact in  $S^1 \times I$ ,

therefore  $f[C]$  is closed, by  $S^1 \times I$  is Hausdorff.

Moreover,  $f(x_1, y_1) = f(x_2, y_2) \Leftrightarrow e^{2\pi i x_1} = e^{2\pi i x_2}$  and  $y_1 = y_2$   
 $(\Leftrightarrow e^{2\pi i(x_1 - x_2)} = 1)$

$$\Leftrightarrow \left( x_1 = x_2 \text{ or } \{x_1, x_2\} = \{0, 1\} \right) \text{ and } y_1 = y_2$$

$$\Leftrightarrow (x_1, y_1) \sim (x_2, y_2)$$

Therefore,  $f$  induces a Homeomorphism

$$\tilde{f}: I^2/\sim \rightarrow S^1 \times I: [(x, y)] \mapsto f(x, y).$$

It is injective and well-defined by  $(x_1, y_1) \sim (x_2, y_2) \Leftrightarrow f(x_1, y_1) = f(x_2, y_2)$

And surjective is given by the  $f$  is surjective.

Finally, the continuity is given by:  $\tilde{f} \circ \pi = f$  is continuous, and,

given closed set  $C \subseteq I^2/\sim$ ,  $\pi^{-1}[C]$  is closed, so the ontoness of  $\pi$  gives

$\tilde{f}[C] = \tilde{f}[\pi[\pi^{-1}[C]]] = f[\pi^{-1}[C]]$  is closed, since  $f$  is closed map.  $\square$

Torus

Def.

Define an equivalence relation  $\sim$  on  $I \times I$  generated by:

$$\begin{cases} \forall x, y \in I, (x, 0) \sim (x, 1) \text{ and } (0, y) \sim (1, y) \\ (0, 0) \sim (1, 0) \sim (0, 1) \sim (1, 1) \end{cases}$$

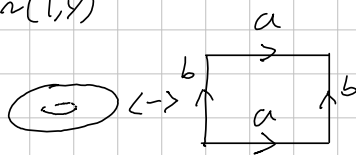
The quotient space  $I^2/\sim$  is called a Torus.

Prop. The Torus  $I^2/\sim$  is Homeomorphic to  $S^1 \times S^1 \subseteq \mathbb{C}$ .

Proof. Define a map

$$f: I^2 \rightarrow S^1 \times S^1: (x, y) \mapsto (e^{2\pi i x}, e^{2\pi i y})$$

Then,  $f$  is continuous onto closed map,



$$\langle a, b \mid aba^{-1}b^{-1} = e \rangle$$

$$\cong \mathbb{Z} \times \mathbb{Z}$$

$$I^2/\sim \cong S^1 \times S^1 \text{ gives}$$

$$\pi_1(I^2/\sim) \cong \pi_1(S^1 \times S^1)$$

$$\cong \pi_1(S^1) \times \pi_1(S^1)$$

Projective space.

Def. Define an equivalence relation on  $\mathbb{R}^{n+1} \setminus \{0\}$  generated by:

$$x \sim y \Leftrightarrow \exists t \in \mathbb{R} \text{ such that } tx = y.$$

The quotient space  $\mathbb{RP}^n \stackrel{\text{def}}{=} (\mathbb{R}^{n+1} \setminus \{0\}) / \sim$  is called a Real Projective space.

Prop.  $\mathbb{RP}^1$  is Homeomorphic to the circle  $S^1$ .

Proof. Define a map

$$f: \mathbb{R}^2 \setminus \{(0,0)\} \rightarrow S^1: (x,y) \mapsto \left( \frac{x}{\sqrt{x^2+y^2}}, \frac{y}{\sqrt{x^2+y^2}} \right)$$

Then, it is continuous clearly, and onto: given  $(\cos \theta, \sin \theta) \in S^1$ , take

$$x = \cos \theta \quad \text{and} \quad y = \sin \theta.$$

Moreover,

$$(x_1, y_1) \sim (x_2, y_2) \Leftrightarrow \exists t \in \mathbb{R} \text{ such that } (x_1, y_1) = (tx_2, ty_2)$$

$$\Leftrightarrow f(x_1, y_1) = f(x_2, y_2)$$

Therefore,  $f$  induces the Homeomorphism

$$\tilde{f}: \mathbb{RP}^1 \rightarrow S^1: [(x,y)] \mapsto f(x,y).$$

So proved.  $\square$

klein bottle.

Möbius band.